University of Ottawa

Department of Mathematics and Statistics

MAT 2143: Algebraic Strutures (Introduction to Group Theory)
Professor: Hadi Salmasian

Midterm Exam

February 27, 2019

Surname _____ First Name _____

Student # _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
(c) Write your student number at the top of each page in the space provided.
(d) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(e) You are strongly recommended to write in **pen**, not pencil.
(f) You may use the last page of the exam as scrap paper.
(g) Cellular phones, unauthorized electronic devices or course notes (unless an open-book exam) are not allowed during this exam. **Phones and devices must be turned off and put away in your bag**. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave immediately the exam and academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.

By signing below, you acknowledge that you have ensured that you are complying with the above statement.

Signature _____

Please do not write in the table below.

Question	1	2	3	4	5	6	7	Total
Maximum	5	3	6	4	2	4	4	28
Grade								

1. (a) **[2 points]** Write down the definition of a group.

Solution: See Definition 2.15 from the notes.

(b) **[3 points]** Write down the proof of the Right Cancellation Law:

Let G be any group, and let $a, b, c \in G$. If $ba = ca$, then $b = c$.

Solution: Assume that $ba = ca$. Multiplying both sides by a^{-1} we obtain

$$ba(a^{-1}) = ca(a^{-1}) \;\Rightarrow\; b(aa^{-1}) = c(aa^{-1}) \;\Rightarrow\; be = ce \;\Rightarrow\; b = c.$$

2. For each of the following sets X with the given binary operation $\star$, determine if $(X, \star)$ is a group or not. You should justify your answers.

(a) **[1.5 points]** $X = \mathbb{Z}$ and $a \star b = ab - a - b$ for all $a, b \in X$.

Solution: (a) X is not a group because there is no identity element. Assume that some $a \in X$ is the identity of X. Then

$$\forall_{b \in X} a \star b = b.$$

Thus $ab - a - b = b$, or equivalently, $b(a-2) = a$. If $a = 2$ then this implies $0 = 2$ which is a contradiction. If $a \neq 2$, then this implies $b = \frac{a}{a-2}$, which is a contradiction because $b \in X$ can be any integer. Thus there is no identity element for $\star$.

(b) **[1.5 points]** $X = \{1, -1, \sqrt{-1}, -\sqrt{-1}\}$ and $a \star b = ab$ for all $a, b \in X$.

Solution: X is a group because it is a subgroup of $\mathbb{C}^* = \mathbb{C} - \{0\}$ equipped with multiplication. This can be checked by the Subgroup Test: X is a subgroup if and only if for all $a, b \in X$ we have $ab^{-1} \in X$. This is easy to verify directly.

3. (a) **[2 points]** Write down the definition of an isomorphism between two groups.

Solution: This is Definition 3.12 of the notes.

(b) For each pair of groups G and H given below, determine if they are isomorphic or not. You should justify your answers.

• **[2 points]** $G = \mathbb{Z}_4 \times \mathbb{Z}_4$ and $H = \mathbb{Z}_{16}$.

Solution: These groups are not isomorphic because $1 \in \mathbb{Z}_{16}$ is an element of order 16, but $\mathbb{Z}_4 \times \mathbb{Z}_4$ does not have any element of order 16. Indeed for $(a, b) \in \mathbb{Z}_4 \times \mathbb{Z} - 4$, we have

$$(a, b)^4 = (a + a + a + a, b + b + b + b) = (0, 0).$$

Thus every element of H has order dividing 4.

Since isomorphisms preserve orders of elements, there can't be an isomorphism between G and H.

• **[2 points]** $G = S_2 \times U(3)$ and $H =$ Group of symmetries of a rectangle with unequal sides.

Solution: We have $S_2 \cong \mathbb{Z}_2$ and $U(2) \cong \mathbb{Z}_2$ because every group of order 2 is isomorphic to $\mathbb{Z}_2$. Thus $G \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Moreover, the group of symmetries of a rectangle with unequal sides contains 4 elemens: the trivial map, reflection about a horizontal line, reflection about a vertical line, and the composition of the latter two reflections. By a straightforward investigation we see that this group is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$. In fact H does not have an element of order 4, and the only groups with 4 elements are $\mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathbb{Z}_4$, and $\mathbb{Z}_4$ has an element of order 4. Thus we can't have $H \cong \mathbb{Z}_4$, so $H \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \cong G$.

4. (a) **[1 point]** Write down the definition of the centre of a group.

Solution: This is Definition 5.10 from the notes.

(b) **[1 point]** Give an example of a group which is not equal to its own centre.

Solution: Any non-abelian group has this property. For example S_n for $n > 2$ has this property.

(c) **[2 points]** Prove the following statement:

Let G be any group. Then the centre of G is an abelian subgroup of G.

Solution: This is Theorem 5.11 from the notes.

5. [**2 points**] For each of the following statements, write **T** if the statement is true, and write **F** if the statement is false. You will receive .5 points for a correct answer, and -.25 points for an incorrect answer. (If you leave the answer space blank, you will not receive any negative points.) You do not need to justify your answers.

____ All infinite cyclic groups are isomorphic to each other.

____ The group of symmetries of a regular pentagon is abelian.

____ If two cycles σ and τ in a symmetric group S_n are disjoint, then their inverses σ^{-1} and τ^{-1} are also disjoint cycles.

____ For every integer $n > 1$, the group $\mathbb{Z}_n$ has a subgroup that is isomorphic to $\mathbb{Z}_p$ for some prime number p.

Solution:
T (all infinite cyclic groups are isomorphic to $\mathbb{Z}$)

F (we know that this Dihedral group is not abelien because $srs^{-1} = r^{-1} = r^4 \neq r$)

T (this follows from Proposition 8.4 in the notes)

T (subgroups of $\mathbb{Z}_n$ are of the form $d\mathbb{Z}_n$ for $d|n$. Such a subgroup is cyclic of order $\frac{n}{d}$. Therefore if we choose $d = \frac{n}{p}$ for some prime divisor p of n, then we obtain that $d\mathbb{Z}_n \cong \mathbb{Z}_p$.)

6. (a) **[2 points]** Write down the cycle decomposition of the permutation $\sigma \in S_{10}$ where

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 3 & 6 & 4 & 9 & 5 & 7 & 2 & 8 & 1 & 10 \end{pmatrix}$$

(b) **[2 points]** Suppose that the permutation $\tau \in S_{12}$ has cycle decomposition

$$\tau = (2\ \ 5\ \ 1\ \ 8)(3\ \ 11\ \ 7)(6\ \ 9).$$

Write down the cycle decomposition of τ^{-1}. You should justify your answer.

Solution: (a) By computation of the orbits of σ we obtain

$$\sigma = (1\ 3\ 4\ 9)(2\ 6\ 7).$$

(b) By Proposition 8.4, we can compute the inverse of each cycle in the cycle decompsition (note that if we write $\tau = \tau_1\tau_2\tau_3$ where τ_1, τ_2, τ_3 are the above cycles then $\tau^{-1} = \tau_1^1\tau_2^{-1}\tau_3^{-1}$ because the cycles τ_1, τ_2, τ_3 commute. Thus:

$$\tau^{-1} = (81\ 5\ 2)(7113)(96).$$

7. (a) **[2 points]** Let G be a group and let $g \in G$ such that $o(g) = 36$. Determine $o(g^{40})$. You should justify your answer.
(b) **[2 points]** Let G be a group and let $g, h \in G$ such that $o(g) = 7$ an $o(h) = 5$. As usual, let $\langle g \rangle$ and $\langle h \rangle$ denote the cyclic subgroups of G generated by g and h, respectively. Set $S := \langle g \rangle \cap \langle h \rangle$. Prove that $|S| = 1$.

Solution: (a) By Theorem 6.8, the order of g^{40} is $\frac{36}{\gcd(36,40)}$. Now $36 = 2^2 \times 3^2$ and $40 = 2^3 \times 5$ and therefore $\gcd(36, 40) = 4$. It follows that $o(g^{40}) = 9$.
(b) Let $x \in \langle g \rangle \cap \langle h \rangle$. We have

$$\in \langle g \rangle \;\Rightarrow\; x = g^k \text{ for some } k, \;\Rightarrow\; o(x) = \frac{7}{\gcd(7,k)} \;\Rightarrow\; o(g)|7.$$

Similarly, from $x \in \langle h \rangle$ it follows that $o(x)|5$. Thus $o(x)$ is a common divisor of 7 and 5, hence $o(x)|\gcd(7,5)$. But $\gcd(7,5) = 1$. It follows that $o(x) = 1$, i.e., $x = e$. Therefore the only element in $\langle g \rangle \cap \langle h \rangle$ is the identity element.